SATYAM AWASTHI

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Research and Systems Engineer specializing at the intersection of **Spatial AI, Autonomous Decision-Making, and Human-Computer Interaction (HCI)**. Proven record of designing published mixed-reality locomotion frameworks and engineering high-throughput, constraint-bound agentic pipelines within large-scale distributed architectures.

EDUCATION

University of California, Santa Barbara — Santa Barbara, CA, US
M.S. in Computer Science (GPA: 3.91/4.00) — 2021 – 2023
Graduate Research focused on Spatial AI, AR/VR, and HCI at the Four Eyes Lab.

Indian Institute of Technology, Kharagpur — Kharagpur, India
Bachelor of Technology in Electrical Engineering, Minor in Computer Science (CGPA: 8.98/10.00) — 2016 – 2020
Graduated with First Class Honors. Completed core navigation pipelines for Autonomous Ground Vehicles.

PUBLICATIONS

- **S. Awasthi**, V. Ross, S. Lim, M. Beyeler, T. Höllerer. "Eye Tracking Performance in Mobile Mixed Reality." *2024 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, 2024. [DOI Link]
- **S. Awasthi**, V. Ross, M. Beyeler, T. Höllerer. "EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion." *2023 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, 2023. [DOI Link]
- S. Das, **S. Awasthi**, A. Shamnan, P. De, S. Chakraborty, B. Mitra. "SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing." *2022 14th International Conference on COMmunication Systems & NETworkS (COMSNETS)*, 2022. [DOI Link]

PROFESSIONAL EXPERIENCE

Intuit Inc — Mountain View, CA
Software Engineer 2 — Mar 2024 – Jan 2026
- Implemented a dual-stage **agentic decision pipeline** leveraging structured LLMs with tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** using RAG to fetch component primitives and a structured LLM to optimize deterministic UI layout configurations.

Yahoo Inc — Mountain View, CA
Software Dev Engineer II (IC3) — Apr 2023 – Nov 2023
- Developed client-side rule engines to process asynchronous receipt payloads, executing deterministic filtering and cost-optimization algorithms for **real-time decision support**.
- Designed a dynamic, state-driven carousel UI framework optimized for high-volume content delivery systems.

Coupang Global LLC — Mountain View, CA
Software Engineering Intern — Jun 2022 – Sep 2022
- Engineered **context-aware filtering heuristics** to prune multi-dimensional search spaces based on user state vectors prior to recommendation execution.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features against pre-filtered user affinity profiles.
- Optimized system performance to increase **request throughput** and reduce **latency scaling curves** while maintaining downstream user conversion baselines.

Samsung Research Institute — Noida, India
Software Engineer (CL 2) — Sep 2020 – Sep 2021
- Acted as Lead Developer for a core system privacy framework, implementing deterministic policy-enforcement algorithms and coordinating integration across 3 platform teams.
- Honored with the Samsung Spot Award for designing verifiably secure architectural patterns and fault-tolerant codebase structures.
- Optimized control logic and finite state machines within the native VisualVoiceMail subsystem to minimize latency and memory overhead under hardware constraints.

SELECTED PROJECTS

JitterNot UCSB, 2022

- Researched and developed targeted signal processing filters and low-latency smoothing algorithms to mitigate noise and jitter in real-time tracking streams.
- Evaluated user interaction metrics across spatial interface layouts to maximize baseline tracking stability and system responsiveness.

In Motion UCSB, 2022

- Designed interactive locomotion and path-navigation modules optimized for immersive tracking modalities within virtual and mixed reality environments.
- Engineered calibration pipelines to align physical movement state vectors smoothly with dynamic, virtual coordinate spaces.

Tweeter: Distributed Infrastructure Testbed UCSB, Fall 2021

- Conducted phasal load testing via Tsung up to 512 users/second, mapping user trajectories as state machines to analyze **latency scaling curves**.
- Eliminated N+1 bottlenecks using database pagination and **ActiveRecord preloading includes** to batch-fetch relational rows via optimized ID collections.
- Optimized high-concurrency read throughput by deploying **nested Russian Doll fragment caches** and horizontal/vertical server auto-scaling.

Autonomous Ground Vehicle & Swarm Robotics IIT Kharagpur, 2016 – 2018

- Built an autonomous navigation stack in ROS, fusing heterogeneous sensor streams (LiDAR, IMU, GPS) via an Extended Kalman Filter (EKF).
- Implemented visual obstacle segmentation networks (SegNet/U-Net) along with global (A*/Dijkstra) and local (DWA) path planners.
- Researched decentralized swarm coordination using peer-to-peer relative pose estimation (AprilTags) over an ad-hoc XBee mesh network.

TECHNICAL SKILLS

Programming Languages: Python, C++, Kotlin, Java, MATLAB, SQL
Frameworks & Core Tools: LangChain, Pydantic (JSON Schema), ROS, Android/iOS Native SDKs, OpenCV, PyTorch, Milvus/Vector Databases, Tsung, ActiveRecord
Methodologies & Core Research: Spatial AI, Autonomous Decision-Making, Human-Computer Interaction (HCI), Agentic Architectures, State-Space Estimation, Distributed Systems Benchmarking, Constraint-Bound Optimization